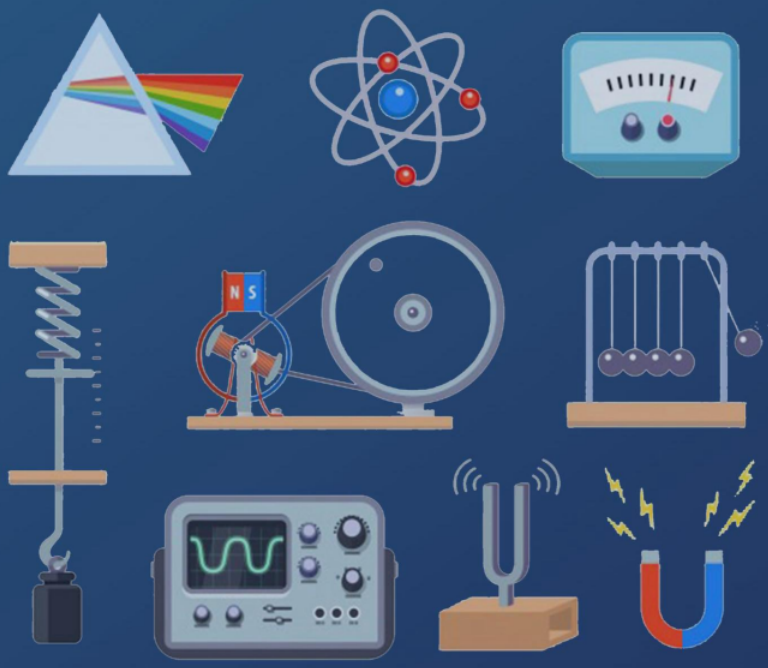
PHYSICS

Radioactivity



SHEET

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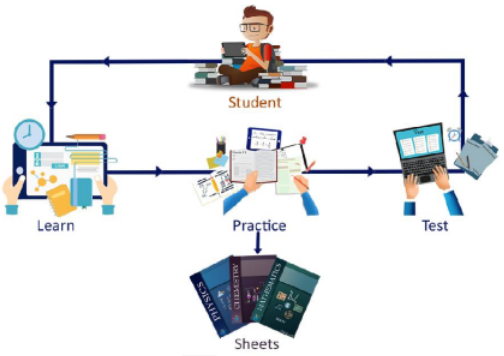
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 3
5.	Exercise 2	Page - 6
6.	Exercise 2A	Page - 8
7.	Answer Key	Page - 12

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Various types of radioactivity Decay	1-7	1-5	1,2	1,19
2	Radioactive Derivation	8-12	6-7,14-16	3-13	2-7,20
3	Activity of a radioactivity sample	13-16	10-12,20	14-16	8,21
4	Application of radioactivity		17	17,18	9,18,22
5	Nuclear fission reaction and nuclear reactor	17-22	8,18,19	19-21	10,23,24
6	Nuclear fusion	23-25		22	
7	After Full Chapter		9,13		11-17

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Various types of radioactive Decay

Q1. Which of the following is true for isotopes of specimen of U^{235} and U^{238} -

(A) both contain same number of neutrons

(B) both contain same of number of proton, electron and neutron

(C) both contain same number of proton and electron but U^{238} contains three more neutrons than U^{235}

(D) U^{238} contain three less neutrons than U^{235}

Q2. Which one of the following pairs of nuclei are isotones :-

(A) $_{34}Se^{74}$, $_{31}Ga^{71}$

(B) $_{42}Mo^{92}$, $_{40}Zr^{92}$

(C) $_{38}Sr^{84}$, $_{38}Sr^{86}$

(D) $_{20}Ca^{40}$, $_{16}S^{32}$

Q3. A nucleus with Z = 92 emits the following in a sequence : $\alpha, \alpha, \beta^-, \beta^-, \alpha, \alpha, \alpha, \alpha, \beta^-, \beta^-, \alpha, \beta^+, \beta^+, \alpha$. The Z of the resulting nucleus is-

(A) 76 (B) 78

(C) 82 (D) 74

Q4. In the uranium radioactive series the initial nucleus is $_{92}U^{238}$ and the final nucleus is $_{82}Pb^{206}$. When the uranium nucleus decays to lead, the number of α,β – particles emitted is-

(A) 6, 8 (B) 8, 6

(C) 16, 6 (D) 32, 12

Q5. When $_3Li^7$ nuclei are bombarded by protons and the resultant nuclei are $_4Be^8$, the emitted particles will be-

(A) alpha particles

(B) beta particles

(C) gamma photons

(D) neutrons

Q6. Which of the following cannot be emitted by radioactive substances during their decay?

(A) Protons (B) Neutrinos

(C) Helium nuclei (D) Electrons

Q7. In gamma ray emission from a nucleus -

(1) both the neutron number and the proton number change

(2) there is no change in the proton number and the neutron number

(3) only the neutron number changes

(4) only the proton number changes

Radioactive Derivation

Q8. The half life of a neutron is 800 sec. 10^8 neutrons at a certain instant are projected from one space station towards another space station, situated 3200 km away with a velocity 2000 m/s. Their velocity remains constant during the journey. How many neutrons reach the other station?

(A) 50×10^6 (B) 25×10^6

(C) 80×10^5 (D) 25×10^5

Q9. A radioactive nuclide can decay simultaneously by two different processes which have decay constants λ_1 and λ_2. The effective decay constant of the nuclide is λ, then :

(A) $\lambda = \lambda_1 + \lambda_2$ (B) $\lambda = 1/2(\lambda_1 + \lambda_1)$

(C) $\frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$ (D) $\lambda = \sqrt{\lambda_1 \lambda_2}$

Q10. The half life of a radioactive element is 30days, in 90days the percentage of disintegrated part is-

(A) 13.5 %
(B) 46.5 %
(C) 87.5%
(D) 90.15%

Q11. 'Rn' decays into 'Po' by emitting α - particle with half life of 4 days, A sample contains 6.4×10^{10} atoms of Rn. After 12 days, the number of atoms of 'Rn' left in the sample will be-

(A) 3.2×10^{10}
(B) 0.53×10^{10}
(C) 2.1×10^{10}
(D) 0.8×10^{10}

Q12. In a radioactive element the fraction of initial amount remaining after its mean life time is-

(A) $1 - \frac{1}{e}$
(B) $\frac{1}{e^2}$
(C) $\frac{1}{e}$
(D) $1 - \frac{1}{e^2}$

Activity of a radioactivity sample

Q13. A radioactive substance is dissolved in a liquid and the solution is heated. The activity of the solution.

(A) is smaller than that of element
(B) is greater than that of element
(C) is equal to that of element
(D) will be smaller or greater depending upon whether the solution is weak or concentrated.

Q14. Activity of a radioactive substance is R_1 at time t_1 and R_2 at time $t_2 (t_2 > t_1)$. Then the ratio $\frac{R_2}{R_1}$ is :

(A) $\frac{t_2}{t_1}$
(B) $e^{-\lambda(t_1+t_2)}$
(C) $e\left(\frac{t_1 - t_2}{\lambda}\right)$
(D) $e^{\lambda(t_1-t_2)}$

Q15. Atomic weight of a radioactive element is M_w gm. Radioactivity of m gm. of its mass is :-

(N_A = Avogadro number, λ = decay constant)

(A) $N_A \lambda$
(B) $\left[\frac{N_A}{M_w} m\right]\lambda$
(C) $\left[\frac{N_A}{m}\right]\lambda$
(D) $\left[\frac{N_A}{m} M_w\right]\lambda$

Q16. What is curie -

(A) measurement of radioacitivty
(B) measurement of temperature
(C) measurement of magnetism
(D) measurement of electric field

Nuclear fission reaction and nuclear reactor

Q17. A nuclear fission is given below

$A^{240} \rightarrow B^{100} + C^{140} + Q(\text{energy})$

Let binding energy per nucleon of nucleus A, B and C is 7.6 MeV, 8.1 MeV and 8.1 MeV respectively. Value of Q is :- (Approximately)

(A) 20 MeV
(B) 220 MeV
(C) 120 MeV
(D) 240 MeV

Q18. In any fission process the ratio $\frac{\text{mass of fission products}}{\text{mass of parent nucleus}}$ is :-

(A) less than 1
(B) greater than 1
(C) equal to 1
(D) depends on the mass of the parent nucleus

Q19. Heavy water is -

(A) Water at 4^0C
(B) Watercontaining various salts
(C) Compound of heavy oxygen and hydrogen
(D) Compound of oxygen and deuterium.

Q20. Nuclear reactor is surrounded by concrete walls to -

(A) Strengthen the construction

(B) Control the chain reaction

(C) form a protective shield

(D) as moderator

Q21. Substance used to slow down the fast neutrons released during nuclear fission is called.

(A) Fuel (B) Moderator

(C) Controlling rods (D) Reflecting rods

Q22. This question contains Statement-1 and Statement-2. Out of the four choices given after the statements, choose the one that best describes the two statements.

Statement-1 : Energy is released when heavy nuclei undergo fission or light nuclei undergo fusion.

Statement-2 : For heavy nuclei, binding energy per nucleon increases with increasing Z while for light nuclei it decreases with increasing Z.

(1) Statement-1 is false, Statement-2 is true.

(2) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

(3) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

(4) Statement-1 is true, Statement-2 is false.

Nuclear fusion

Q23. Solar energy is mainly due to :-

(A) burning of hydrogen in the oxygen

(B) fission of uranium present in the Sun

(C) fusion of protons during synthesis of heavier elements

(D) gravitational contraction

Q24. The following nuclear reaction is :-

$4\ {}_1H^1 \rightarrow {}_2He^4 + 2\ {}_1e^0 + 2\ \nu + 26MeV$

(A) fission

(B) fusion

(C) disintigration

(D) scattering of α–particle

Q25. Inside the sun -

(A) Four nuclei of hydrogen combine to form two nuclei of helium.

(B) Four nuclei of hydrogen combine to form four nuclei of helium.

(C) Four nuclei of hydrogen combine to form one nucleus of helium.

(D) Four nuclei of hydrogen is transformed into one nucleus of helium.

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following is/are incorrect statements ?

(A) Beta rays are same as cathode rays

(B) Gamma rays are high energy neutrons

(C) Alpha particles are singly ionized helium atoms

(D) Protons and neutrons have exactly the same mass

Q2. A nucleus undergoes a series of decay according to the scheme

$$A \xrightarrow{\alpha} B \xrightarrow{\beta} C \xrightarrow{\alpha} D \xrightarrow{\gamma} E$$

Atomic number and mass numbers of E are 69 and 172.

(A) Atomic number of A is 72

(B) Mass number of B is 176

(C) Atomic number of D is 69

(D) Atomic number of C is 69

Q3. The instability of the nucleus can be due to various causes. An unstable nucleus emits radiations if possible to transform into less unstable state. Then the cause and the result can be

(A) a nucleus of excess nucleons is α active

(B) an excited nucleus of excess protons is β^- active

(C) an excited nucleus of excess protons is β^+ active

(D) an nucleus of excess neutrons is β^- active

Q4. When a nucleus with atomic number Z and mass number A undergoes a radioactive decay process:

(A) both Z and A will decrease, if the process is α decay

(B) Z will decrease but A will not change, if the process is β^+ decay

(C) Z will decrease but A will not change, if the process is β^- decay

(D) Z and A will remain unchanged, if the process is γ decay.

Q5. When the mass number A of the nucleus increases

(A) initially the neutron–proton ratio is constant = 1

(B) initially neutron–proton ratio increases and later decreases

(C) initially binding energy per nucleon increases and later decreases

(D) the binding energy per nucleon increases when the neutron–proton ratio increases.

Q6. Half life for certain radioactive element is 5 min. Four nuclei of that element are observed at a certain instant of time. After five minutes :

Assertion (A) : It can be definitely said that two nuclei will be left undecayed

Reasoning (R) : After half life i.e. 5 minutes, half of total nuclei will disintegrate. So only two nuclei will be left undecayed. Then -

(A) A is correct & R is correct explanation of A

(B) Both are correct. But R is not correct explanation of A

(C) A is incorrect & R is correct

(D) Both are incorrect

Q7. The decay constant of the end product of a radioactive series is

(A) zero

(B) infinite

(C) finite (non zero)

(D) depends on the end product

Q8. Which of the following is correct for a nuclear reaction?

(A) A typical fission represented by

$${}_{92}U^{235} + {}_{0}n^{1} \rightarrow {}_{56}Ba^{143} + {}_{36}Kr^{93} + \text{energy}$$

(B) Heavy water is used as moderator in preference to ordinary water because H may capture neutrons, while D would not.

(C) Cadmium rods increase the reactor power when they go in, decrease when they go outward.

(D) Slower neutrons are more effective in causing fission than faster neutrons in case of U^{235}.

Q9. When the nucleus of an electrically neutral atom undergoes a radioactive decay process, it will remain neutral after the decay if the process is :

(A) α decay (B) β^- decay

(C) γ decay (D) K-capture

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question. 10 to 12

Many unstable nuclei can decay spontaneously to a nucleus of lower mass but different combination of nucleons. The process of spontaneous emission of radiation is called radioactivity. Three types of radiations are emitted by radioactive substance. Radioactive decay is a statistical process. Radioactivity is indepndent of all external conditions.

The number of decays per unit time or decay rate is called activity. Activitiy exponentially decreases with time. Mean lifetime is always greater than half-life time.

Q10. Choose the correct statement about radio-activity.

(A) Radioactivity is a statistical process

(B) Radioactivity is independent of high temperature and high pressure

(C) When a nucleus undergoes $\alpha-$ or $\beta-$ decay, its atomic number changes

(D) All of these

Q11. If T_H is the half-life and T_M is the mean life. Which of the following statements is correct:

(A) $T_M > T_H$

(B) $T_M < T_H$

(C) Both are directly proportional to square of the decay constant

(D) $T_M \propto \lambda_0$

Q12. 'n' number of α particles per second are being emitted by N atoms of a radioactive element. The half-life of element will be :

(A) $\frac{n}{N}s$ (B) $\frac{N}{n}s$

(C) $\frac{0.693N}{n}s$ (D) $\frac{0.693n}{N}s$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q13. Match the following Columns :-

Column 1	**Column 2**
(A) Nuclear fusion	(P) Converts some matter into energy
(B) Nuclear fission	(Q) Generally occurs for nuclei with low atomic number
(C) β–decay	(R) Generally occurs for nuclei with higher atomic number
(D) Exothermic nuclear reaction	(S) Essentially proceeds by weak nuclear forces

SUBJECTIVE TYPE

Q14. A radioactive nucleus undergoes a series of decays according to the following scheme:

$A \xrightarrow{\alpha} A_1 \xrightarrow{\beta^-} A_2 \xrightarrow{\alpha} A_3 \xrightarrow{\gamma} A_4$

If the mass number and atomic number of A are 180 and 72 respectively, what are these numbers for A_4?

Q15. The half lives of radioactive elements x and y are 3 minute and 27 minute respectively. If the activities of both are same, then calculate the ratio of number of atoms of x and y.

Q16. $\frac{2}{3}$ fraction of a sample disintegrates in 7 days. How much fraction of it will decay in 21 days?

Q17. The half life of a radioactive material is T. After T/2 time, the material left is -

Q18. ${}_{92}U^{238} \xrightarrow{\alpha} X_1 \xrightarrow{+\beta^0} {}_aX^b$, find a & b.

Q19. ${}_{92}U^{238} \xrightarrow{n\alpha, n'_\beta^0} {}_{82}Pb^{206}$. Find n & n'

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q20. After 280 days, the activity of a radioactive sample is 6000 dps. The activity reduces to 3000 dps after another 140 days. The initial activity of the sample in dps is :

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. ${}_{92}U^{235}$, nucleus absorbs a neutron and disintigrate in ${}_{54}Xe^{139}$, ${}_{38}Sr^{94}$ and X so, What will be product 'X' :-

(A) 3 - neutrons (B) 2-neutrons
(C) α - particle (D) β - particle

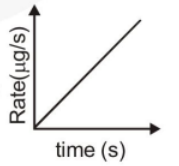
Q2. The number of α and β^- emitted during the radioactive decay chain starting from $^{226}_{88}Ra$ and ending at $^{206}_{82}Pb$ is -

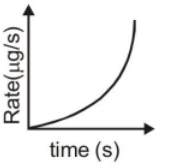
(A) 3α & $6\beta^-$ (B) 4α & $5\beta^-$
(C) 5α & $4\beta^-$ (D) 6α & $6\beta^-$

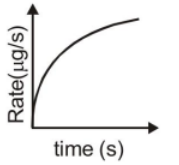
Q3. Initial ratio of active nuclei in two different samples is 2:3. Their half lives are 2hr and 3hr respectively. Ratio of their activities at the end of 12hr is:-

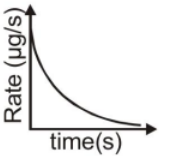
(A) 1 : 6 (B) 6 : 1
(C) 1 : 4 (D) 4 : 1

Q4. The radioactive nuclide of an element X decays to a stable nuclide of element Y. Then, in a given sample of X, the rate of formation of Y is given by the graph -

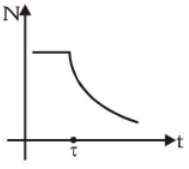
(A) 

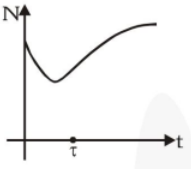
(B) 

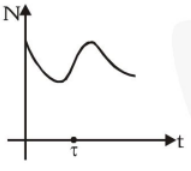
(C) 

(D) 

Q5. A radioactive sample consists of two distinct species having equal number of atoms initially. The mean life of one species is τ and that of the other is 5τ. The decay products in both cases are stable. A plot is made of the total number of radioactive nuclei as a function of time. Which of the following figure best represent the form of this plot ?

(A) 

(B) 

(C) 

(D) 

Q6. The half–life of ^{131}I is 8 days. Given a sample of ^{131}I at time t = 0, we can assert that :

(A) No nucleus will decay before t = 4 days

(B) No nucleus will decay before t = 8 days

(C) All nuclei will decay before t = 16 days

(D) A given nucleus may decay at any time after t = 0

Q7. The half–life period of a radioactive element x is same as the mean life time of another radioactive element y. Initially both of them have the same number of atoms. Then :

(A) x and y have the same decay rate initially

(B) x and y decay at the same rate always

(C) y will decay at a faster rate than x

(D) x will decay at faster rate than y

Q8. Two radioactive material A_1 and A_2 have decay constants of $10\lambda_0$ and λ_0. If initially they have same number of nuclei, the ratio of number of their undecayed nuclei will be (1/e) after a time

(A) $\frac{1}{\lambda_0}$ (B) $\frac{1}{9\lambda_0}$ (C) $\frac{1}{10\lambda_0}$ (D) 1

Q9. 90% of a radioactive sample is left undecayed after time t has elapsed. What percentage of the initial sample will decay in a total time 2t :

(A) 20% (B) 19%

(C) 40% (D) 38%

Q10. Starting with a sample of pure ^{66}Cu, 7/8 of it decays into Zn in 15 min. The corresponding half-life is-

(1) 10 min (2) 15 min

(3) 5 min (4) $7\frac{1}{2}$ min

Q11. If $N_t = N_0e^{-\lambda t}$ then number of disintegrated atoms between t_1 to t_2 $(t_2 > t_1)$ will be :–

(1) $N_0\left[e^{\lambda t_2} - e^{\lambda t_1}\right]$

(2) $N_0\left[-e^{\lambda t_2} - e^{-\lambda t_1}\right]$

(3) $N_0\left[e^{-\lambda t_1} - e^{-\lambda t_2}\right]$

(4) None of these

Q12. A sample of radioactive element has a mass of 10 gm at an instant t = 0. The active mass of this element in the sample after two mean lives is :-

(A) 1.35 gm (B) 2.50 gm

(C) 3.70 gm (D) 6.30 gm

Q13. The half life of a radioactive element is 30 days, in 90 days the percentage of disintegrated part is :–

(A) 13.5 % (B) 46.5 %

(C) 87.5 % (D) 90.15 %

Q14. The activity of a radioactive sample is 9750 counts/min at t = 0 and 975 counts/min at t = 5 minute. The decay constant is appox :-

(A) 0.922 min^{-1} (B) 0.691 min^{-1}

(C) 0.461 min^{-1} (D) 0.230 min^{-1}

Q15. The activity of a sample reduces from A_0 to $A_0/\sqrt{3}$ in one hour. The activity after 3 hours more will be :

(A) $\frac{A_0}{3\sqrt{3}}$ (B) $\frac{A_0}{9}$ (C) $\frac{A_0}{9\sqrt{3}}$ (D) $\frac{A_0}{27}$

Q16. A radioactive material of half-life T was produced in a nuclear reactor at different instants, the quantity produced second time was twice of that produced first time. If now their present activities are A_1 and A_2 respectively then their age difference equals:

(A) $\frac{T}{\ln 2}\left|\ln\frac{A_1}{A_2}\right|$ (B) $T\left|\ln\frac{A_1}{A_2}\right|$

(C) $\frac{T}{\ln 2}\left|\ln\frac{A_2}{2A_1}\right|$ (D) $T\left|\ln\frac{A_2}{2A_1}\right|$

Q17. A particular nucleus in a large population of identical radioactive nuclei did survive 5 half lives of that isotope. Then the probability that this surviving nucleus will survive the next half life :

(A) $\frac{1}{32}$ (B) $\frac{1}{5}$ (C) $\frac{1}{2}$ (D) $\frac{1}{10}$

Q18. A certain radio active substance has a half life of 5 years. Thus for a nucleus in a sample of the element, the probability of decay in ten years is -

(A) 50% (B) 75%

(C) 100% (D) 60%

Q19. Energy released by 1 kg. U^{235}, when it is fissioned is? Assume 200MeV of energy is released per fission of $_{92}U^{235}$ atom.

(A) 8×10^{10} KWH (B) 5×10^{30} eV

(C) 10^{10} Joule (D) 5×10^{26} MeV

Q20. Assuming that 200 MeV of energy is released per fission of $_{92}U^{235}$ atom. Find the number of fission per second required to release1 kW power :-

(A) 3.125×10^{13} (B) 3.125×10^{14}

(C) 3.125×10^{15} (D) 3.125×10^{16}

Q21. In a certain nuclear reactor, a radioactive nucleus is being produced at a constant rate = 1000 /s. The mean life of the radionuclide is 40 minutes. At steady state, the number of radionuclide will be :

(A) 4×10^4 (B) 24×10^4

(C) 24×10^5 (D) 24×10^6

Q22. In the nuclear fusion reaction,

$$_1^2H + _1^3H \rightarrow _2^4He + n$$

given that the repulsive potential energy between the two nuclei is 7.7×10^{-14} J, the temperature at which the gases must be heated to initiate the reaction is nearly [Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K]-

(A) 10^7 K (B) 10^5 K

(C) 10^3 K (D) 10^9 K

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A star initially has 10^{40} deuterons. It produces energy via the processes $_1H^2 + _1H^2 \rightarrow _1H^3 +$ p and $_1H^2 + _1H^3 \rightarrow _2He^4 +$ n. If the average power radiated by the star is 10^{16} W, the deuteron supply of the star is exhausted in a time of the order of :

(A) 10^6 s (B) 10^8 s (C) 10^{12} s (D) 10^{16} s

The mass of the nuclei are as follows :

$M(H^2) = 2.014$ amu ; $M(n) = 1.008$ amu ; $M(p) = 1.0087$ amu ; $M(He^4) = 4.001$ amu.

Q2. The half–life of ^{215}At is 100 μs. The time taken for the radioactivity of a sample of ^{215}At to decay to $1/16^{th}$ of its initial value is :

(A) 400μs (B) 63μs

(C) 40μs (D) 300μs

Q3. The half life of a radioactive substance is T_0. At t=0, the number of active nuclei are N_0. Select the correct alternative.

(A) The number of nuclei decayed in time internal 0–t is $N_0e^{-\lambda t}$

(B) The number of nuclei decayed in time interval 0–t is $N_0(1-e^{-\lambda t})$

(C) The probability that a radioactive nuclei does not decay in interval 0–t is $e^{-\lambda t}$

(D) The probability that a radioactive nuclei does not decay in interval $1-e^{-\lambda t}$

Q4. 7/8 fraction of a sample disintegrates in t time. How much time it will take to disintegrete 15/16 fraction :-

(A) t (B) $\frac{4}{3}t$

(C) $\frac{5}{3}t$ (D) 2t

Q5. A radioactive material decays by simultaneous emission of two particles with respective half-lives 1620 and 810 years. The time (in years) after which one–fourth of the material remains is :-

(A) 1080 (B) 2430

(C) 3240 (D) 4860

Q6. The decay constant of a radio active substance is 0.173 $(years)^{-1}$. Therefore :

(A) Nearly 63% of the radioactive substance will decay in (1/0.173) year.

(B) half life of the radio active substance is (1/0.173) year.

(C) one -forth of the radioactive substance will be left after nearly 8 years.

(D) all the above statements are true.

Q7. At time t = 0, N_1 nuclei of decay constant λ_1 & N_2 nuclei of decay constant λ_2 are mixed. The decay rate of the mixture is :

(A) $N_1N_2e^{-(\lambda_1+\lambda_2)t}$

(B) $+\left(\frac{N_1}{N_2}\right)e^{-(\lambda_1-\lambda_2)t}$

(C) $+\left(N_1\lambda_1e^{-\lambda_1t}+N_2\lambda_2e^{-\lambda_2t}\right)$

(D) $+N_1\lambda_1\,N_2\lambda_2\,e^{-(\lambda_1+\lambda_2)t}$

Q8. A radioactive sample S_1 having an activity of 5μCi has twice the number of nuclei as another sample S_2 which has an activity of 10μCi. The half lives of S_1 and S_2 can be :

(A) 20 years and 5 years, respectively

(B) 20 years and 10 years, respectively

(C) 10 years each

(D) 5 years each

Q9. The half-life of substance X is 45 years, and it decomposes to substance Y. A sample from a meteorite was taken which contained 2% of X and 14% of Y by quantity of substance. If substance Y is not normally found on a meteorite, what is the approximate age of the meteorite?

(A) 270 years (B) 135 years

(C) 90 years (D) 45 years

Q10. Which of the following statement(s) is/are correct?

(A) The rest mass of a stable nucleus is less than the sum of the rest masses of its separated nucleons.

(B) The rest mass of a stable nucleus is greater than the sum of the rest masses of its separated nucleons.

(C) In nuclear fusion, energy is released by fusion of two nuclei of medium mass (approx. 100 amu)

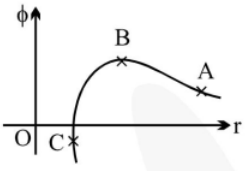
(D) In nuclear fission, energy is released by fragmentation of a very heavy nucleus.

Q11. In an α-decay the kinetic energy of α-particle is 48 MeV and Q-value of the reaction is 50MeV. The mass number of the mother nucleus is : (Assume that daughter nucleus is in ground state)

(A) 96 (B) 100

(C) 104 (D) None of these

Q12. The graph shown by the side shows the variation of potential energy ϕ of a proton with its distance 'r' from a fixed sodium nucleus, as it approaches the nucleus, placed at origin O. Then the portion.



(A) AB indicates nuclear repulsion

(B) AB indicates electrostatic repulsion

(C) BC indicates nuclear attraction

(D) BC represents electrostatic interaction

Q13. In β-decay, the Q-value of the process is E. Then

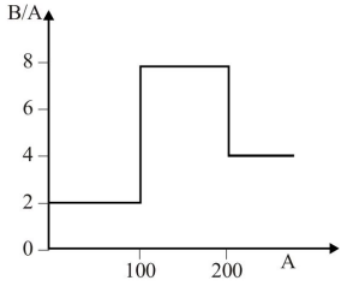
(A) K.E. of a β-particle cannot exceed E.

(B) K.E. of anti neutrino emitted lies between Zero and E.

(C) N/Z ratio of the nucleus is altered.

(D) Mass number (A) of the nucleus is altered.

Q14. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Use this plot to choose the correct choice(s) given below :



(A) Fusion of two nuclei with mass numbers lying in the range of $1 < A < 50$ will release energy.

(B) Fusion of two nuclei with mass numbers lying in the range of $51 < A < 100$ will release energy.

(C) Fission of a nucleus lying in the mass range of $100 < A < 200$ will release energy when broken into two equal fragments.

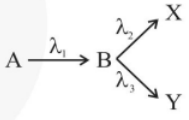
(D) Fission of a nucleus lying in the mass range of $200 < A < 260$ will release energy when broken into two equal fragments.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question Nos. 15 to 17

The rate at which a particular decay process occurs in a radio active sample, is proportional to the number of radio active nuclei present. If N is the number of radio active nuclei present at some instant, the rate of change of N is $\frac{dN}{dt} = -\lambda N$



Consider radioactive decay of A to B which further decays either to X or Y, λ_1, λ_2 and λ_3 are decay constants for A to B decay, B to X decay and Bto Y decay respectively. At $t = 0$, the num,ber of nuclei of A,B,X and Y are N_0, N_0, zero and zero respectively. N_1, N_2, N_3 and N_4 are the number of nuclei of A,B,X and Y at any instant t.

Q15. The net rate of accumulation of B at any instant is -

(A) $N_1\lambda_1 + N_2\lambda_2 + N_3\lambda_3$

(B) $N_1\lambda_1 - N_3\lambda_2 - N_4\lambda_3$

(C) $N_1\lambda_1 - N_2\lambda_2 - N_2\lambda_3$

(D) $N_1\lambda_1 + N_2\lambda_2 - N_2\lambda_3$

Q16. The number of nuclei of B will first increase and then after a maximum value, it decreases for

(A) $\lambda_1 > \lambda_2 + \lambda_3$

(B) $\lambda_1 = \lambda_2 = \lambda_3$

(C) $\lambda_1 = \lambda_2 + \lambda_3$

(D) For any values of λ_1, λ_2 and λ_3

Q17. At $t = \infty$, which of following is incorrect?

(A) $N_2 = 0$

(B) $N_3 = \dfrac{N_0 \lambda_2}{\lambda_2 + \lambda_3}$

(C) $N_4 = \dfrac{2N_0 \lambda_3}{\lambda_2 + \lambda_3}$

(D) $N_1 + N_2 + N_3 + N_4 = 2N_0$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q18. A wooden piece of great antiquity weighs 50 gm and shows C^{14} activity of 320 disintegrations per minute. Estimate the length of the time in years which has elapsed since this wood was part of living tree, assuming that living plants show a C^{14} activity of 12 disintegrations per minute per gm. The half life of C^{14} is 5730 yrs.

SUBJECTIVE TYPE

Q19. A radioactive nucleus ${}_zX^A$ emite 3α–particles and 5β–particles. The ratio of number of neutrons to that of protons in the product nucleus will be :–

Q20. An experiment is done to determine the half–life of radioactive substance that emits one β–particle for each decay process. Measurement show that 8.4 β are emitted each second by 2.5 mg of the substance. The atomic weight of the substance is 230. Find the half life of the substance.

Q21. The activity of a sample of radioactive material is A_1 at time t_1 and A_2 at time t_2 ($t_2 > t_1$). Its mean life is T. Find Az.

Q22. U^{238} and U^{235} occur in nature in an atomic ratio 140 : 1. Assuming that at the time of earth's formation the two isotopes were present in equal amounts. Calculate the age of the earth.

(Half life of $u^{238} = 4.5 \times 10^9$ yrs & that of $U^{235} = 7.13 \times 10^8$ yrs)

Q23. 200 MeV of energy can be obtained by per fission. In a reactor generating 1000 kW find the number of nuclei under going the fission per second :-

Q24. The amount of U^{235} to be fissioned, to operate 10 kW nuclear reactor is (Approximately). Assume 200 MeV of energy can be obtained by per fission.

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	A	B	B	C	A	B	B	A	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	C	C	D	B	A	C	A	D	C
21	**22**	**23**	**24**	**25**					
B	D	C	B	C					

Ex-1A

1. B,C,D **2.** B,C **3.** A, C, D **4.** A, B, D **5.** A, C

6. D **7.** A **8.** A,B,D **9.** C **10.** D

11. A **12.** C

13. (A) P, Q; (B) P, R; (C) P, S ; (D) P, Q, R

14. Mass number = 172 and Atomic number = 69

15. $\frac{1}{9}$ **16.** $\frac{26}{27}$ **17.** $\frac{1}{\sqrt{2}}$ **18.** a = 89, b = 234

19. n = 8, n' = 6 **20.** 24000

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	C	C	D	D	D	C	B	B	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	A	C	C	B	C	C	B	D	A
21	**22**								
C	D								

Ex-2A

1. C **2.** A **3.** B,C **4.** B **5.** A

6. A,C **7.** C **8.** A **9.** B **10.** A,D

11. B **12.** B,C **13.** A,B,C **14.** B,D **15.** C

16. A **17.** B **18.** 5196 **19.** $\frac{A-Z-11}{Z-1}$

20. 1.7×10^{10} years **21.** $A_2 = A_1 e^{(t_1-t_2)/T}$ **22.** 6.04×10^9 yrs

23. 3.125×10^{16} **24.** 1.2×10^{-7} gm/s